Task 4: Stochastic Calculations

We want to examine the probability distributions (variance as well as shape) of our histograms. To do so, we need to figure out the probability distribution of the particles over time.

I think it's easy enough to see that the initial distribution of particles $\frac{1}{N}n(x, 0)$ is the initial probability distribution, since that's how I initialize them by definition. However, it's also the case that at any time t, $\frac{1}{N}n(x,t)$ is the probability distribution of the particles. This is because this distribution is ultimately the sum of the distributions of all our particles, and averages them out, however all of these distributions for our individual particles obey the heat equation. So, because the heat equation is linear, this sum of probability distributions also obeys it. So our probability density function over time looks like some function whose initial conditions are $\frac{1}{N}n(x,0)$ and who follows the heat equation, for which there's only one solution: $\frac{1}{N}n(x,t)$.

Let's now use this to see how our bins are affected by this probability. Consider a bin centered at x_0, with width Δx at time t. The chance that any given particle falls in that bin at time t is a constant probability given by our probability density function:

$$p = \frac{1}{N}\int_{x_0}^{x_0+\Delta x} n(x,t)\,dx \tag{1}$$

If we assume Δx is sufficiently small, the integral simplifies to:

$$p = \frac{1}{N}n(x,t)\Delta x \tag{2}$$

Since the individual particles are independent of each other, we have a binomial distribution. This makes things simple as there are known formulas for the mean and standard deviation:

$$\mu = Np = n(x,t)\Delta x \tag{3}$$

$$\sigma = \sqrt{Np(1-p)} = \sqrt{n(x,t)\Delta x\left(1 - \frac{1}{N}n(x,t)\Delta x\right)} \tag{4}$$

These formulas give us the number of particles however. If we want number density, we need to divide both by Δx:

$$\mu_n = n(x,t) \tag{5}$$

$$\sigma_n = \sqrt{\frac{n(x,t)}{\Delta x}\left(1 - \frac{1}{N}n(x,t)\Delta x\right)} \tag{6}$$

At this point we can also plug in specific $n(x,t)$ to get specific values. Let's do a uniform distribution, where $n(x,t) = N/L$ for N particles in a periodic space of size L:

$$\mu_n = N/L \tag{7}$$

$$\sigma_n = \sqrt{\frac{N}{L\Delta x}\left(1 - \frac{1}{N}\frac{N}{L}\Delta x\right)} \tag{8}$$

$$\sigma_n = \sqrt{\frac{N}{L\Delta x}\left(1 - \frac{\Delta x}{L}\right)} \tag{9}$$

Plotting this against the actual data: